

Willow Memory Security & Governance Playbook (2026)

This document defines the mandatory security architecture, implementation plan, and verification strategy for Willow's autonomous agent ecosystem. It prevents accidental or adversarial destruction of semantic memory (AuraDB / Neo4j) while preserving autonomy, experimentation, and cross-project knowledge transfer.

1) Threat Model

Agents can generate arbitrary Cypher at machine speed.
Therefore the system must be secure-by-construction:

- No raw DB access
- No arbitrary Cypher
- No destructive writes

2) NonNegotiable Security Invariants

1	No agent possesses AuraDB credentials.
2	No agent can submit arbitrary Cypher.
3	All agent writes are append-only.
4	Canonical memory cannot be mutated by agents.
5	Destructive actions require two-phase commit.
6	Every write includes provenance metadata.

3) Target Architecture: Graph Gateway

Agents → Graph Gateway API → AuraDB
Agents never see DB credentials or Cypher.

4) Implementation Phases

Phase 0: Remove direct DB access
Phase 1: Build Gateway
Phase 2: Append-only memory
Phase 3: Monitoring + circuit breakers
Phase 4: Knowledge Packages

5) Verification Plan

Negative tests: DELETE, DROP, APOC
Abuse tests: high write rate
Positive tests: valid episode writes